Project

(ASW Spread on BTPs and BONOs)

Financial Engineering

It is required to construct the ASW-Spread curve versus EONIA for Italian BTPs and Spanish Bonos in the time interval between t_1=01/01/2007 a t_N=31/12/2015.

The dataset, with data from Bloomberg info provider, includes for each date in the time interval:

a. OIS (i.e. Overnight Index Swap) rates with different expiries;

b. BTP and BONOS main information and clean prices (Pls do not consider dirty prices values in the file for BTPs).

It is required to reply to the following points.

A. Build a database

1. Read the excel file & bootstrap the EONIA discount curve using OIS rates up to 10y (see e.g. [1]) for each business day in the time interval considered.
2. Read the excel dataset and select only bonds with fixed coupon (and exclude e.g. Inflation linked ones, Zero Coupons, Floaters ...) and settle date is in the range [01/01/1999, t_N].
3. Filter the data as discussed in [2]. Mention in the document each activated filter.
4. Save a .mat file (in matlab) with:
    a. a vector "bond" (with length equal to the number of bonds that satisfy the criteria) of struct with fields (some fields are vectors theirselves):

       i) BBGname ii) settleDate iii) expDate iv) firstCouponDate v) couponValue vi) couponFrequency vii) pricesDates viii) pricesCleanValues ix) pricesDirtyValues;

       i.e. the ith element will be

       bond(i).BBGname

       bond(i).settleDate, etc.

    b. a vector "EONIA" (with length equal to the number of business days that satisfy the criteria) with three vector fields each

       i) Dates (i.e. settlement date and the expiries of OIS contracts used in the boootstrap), ii) Rates (OIS rates) iii) DiscountFactors. For the OIS bootstrap follow e.g. [1].

B. Compute ASW spread over EONIA and Zeta spread over EONIA for the two countries

0. Observe qualitatively the ASW curve and the Zeta spread curve. Can you identify some "critical" days in these curves for the two countries during the 2011-2012 debt crisis?

1. For every business day t_i between t_1 and t_N select available bonds with expiry in the range [t_i + 2m, t_i + 10y]

2. For every selected bond compute the ASW spread over EONIA as in [2]. The set of floating legs is determined quarterly, modified following, starting backwards from the expiry. The quarterly floating leg (daycount: Act/360) is equal to the corresponding 3m OIS that "fixes" at the beginning of each period. The first one is, in general, a short stub and the corresponding rate is established from the EONIA Discounting curve [1].

3. Save a .mat file with:

    a. two vectors of Spreads (one for each country) with three vector fields as components for each date:

       i) ExpiryDates (i.e. bonds' expiry dates), ii) ASWSpreads, iii) ZetaSpreads.

C. Construct the best fit of ASW Spreads for a straight line broken in one point

1. Filter out the months as discussed in [2] (i.e. when ASW spread is constantly below 20 bps).

2. Compute, on the remaining dates, the best fit of the ASW spread with a straight line broken in one point as in [2] (see Appendix). As in [2] it is required to have in each segment at least 3 points.

D. Reconstruct the Financial Stress Index with the ASW as in [2]. Discuss the results.

E. Reconstruct the Financial Stress Index with the Zeta Spread (Repeat for the Zeta Spread what has been done in points C. and D. for the ASW spread). Discuss the results.

Realize a library in Matlab.

[1] R. Baviera and A. Cassaro (2015), A Note on Dual-Curve Construction: Mr. Crab's Bootstrap, Applied Mathematical Finance 22 (2015), 105-132.

[2] R. Baviera and D. Lebovitz (2015), A sovereign bond based index for financial instability in the euro zone and the role of carry trade, Mimeo

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